

1. Innate immune system is _____.

- (a) Antigen depending
- (b) Specific
- (c) Non specific
- (d) Need activated

2. Which of the cell's play role in tumor surveillance?

- (a) Neutrophils
- (b) Macrophages
- (c) Natural killer NK cells
- (d) Eosinophils

3. Lymph nodes and Peyer's patches are

- (a) Primary lymphoid organ
- (b) Secondary lymphoid organ
- (c) Generative lymphoid organ
- (d) Stem cells

4. Dendritic cells belongs to _____.

- (a) T lymphocytes
- (b) B lymphocytes
- (c) Lymphoid lineage
- (d) Myeloid lineage

5. Primary defense against pathogens is _____.

- (a) Secretory immune system
- (b) B lymphocytes

(c) Cell mediated immune response

(d) Complement activation

6. The portion of antigen combine with immune system called_____.

(a) Immunogen

(b) Hapten

(c) Epitope

(d) Carrier molecule

7. Hypervariable region and framework region are sub-region of_____.

(a) Hinge region

(b) Disulfide bond

(c) Variable region

(d) Constant region

8. Particulate region of antigen cause the reaction of antibody to form clump called_____.

(a) Precipitation

(b) Agglutination

(c) Complement fixation

(d) Immunoprecipitation

9. Antigen and antibody reaction which can be detected on the basis of ability to fix or consume complement is _____.

(a) Radial immunodiffusion

(b) Agglutination

(c) Complement fixation

(d) Elisa

10. Complement proteins are produced by _____

(a) Hepatocytes

- (b) Macrophages
- (c) Gut epithelial cells
- (d) All of these

11. Pro kinin is activated by serum factor called_____.

- (a) Cytokine
- (b) Plasmin
- (c) Opsonin

(d) Anaphylatoxin

12. Soluble immune complex is formed in_____ Hypersensitivity.

- (a) Type 1
- (b) Type 2

(c) Type 3

- (d) Type 4

13. Delayed type of hypersensitivity is transfer with_____.

- (a) B cells

(b) T cells

- (c) Antibody
- (d) Antigen

14. Immune evasion occurs_____.

(a) By weak immune response

- (b) By strong immune
- (c) By strategies
- (d) By both

15. Cell mediated immune response is primarily defense against ____ pathogens.

(a) Intracellular

(b) Extra cellular

(c) Both

(d) None

16. Which of the following is opsonin?

(a) IgG

(b) IgM

(c) Complement proteins

(d) All of these

17. Pyroptosis is type of death induced after _____.

(a) Phagocytosis

(b) Inflammasomes

(c) Complement fixation

(d) Chemotaxis

18. Tolerance of _____ is longer than b cells

(a) T cells

(b) Nk cells

(c) Macrophages

(d) Leucocytes

19. Person have immunodeficiency called _____.

(a) Opportunistic

(b) Healthy

(c) Immunocompromised

(d) Disease

20. Hypo-complementemia is_____.

(a) Abnormalities in complement proteins

(b) Abnormalities in the cells

(c) Abnormalities in NK cells

(d) Abnormalities in b cells

21. Animal with hybrid nature of immune cells with host and donor cells are_____.

(a) Chimeras

(b) Twin

(c) Donor

(d) Host

22. Complement fixation is not carried out by _____ molecule.

(a) IgE Not Sure

(b) IgG

(c) IgM

(d) IgA

23. There are ___ variable gene for lambda light chain.

(a) 20

(b) 4

(c) 30

(d) 40

24. Pneumococcus polysaccharide serve as _____ antigen.

(a) Poor

(b) T dependent

(c) T independent

(d) Monomeric

25. IL-4 involve in the class switching of antibody into_____.

(a) IgA

(b) IgG

(c) IgM

(d) IgE

26. All are live attenuated vaccine except_____.

(a) Small pox

(b) Polio

(c) Measles

(d) Hepatitis A virus

27. Patient suffering AIDS are susceptible to_____.

(a) Neuroblastomas

(b) Melanomas

(c) Lymphoma

(d) Kaposi sarcoma

28. Endogenous antigen residing in_____

(a) Mitochondria

(b) Chloroplast

(c) Nucleus

(d) Cytosol Not Sure

Qno1: What are the examples of endogenous tissue? 3 Marks

Example of endogenous antigen:

Erythroblastosis Fetalis

- Also called as Hemolytic disease of newborn (HDN)
- Rh incompatibility B/W mother & fetus

Other Examples of endogenous antigen:

Autoimmune Tissues disease

- Goodpasture's Syndrome:

Autoantibodies against glomerulus basement membrane

Qno2: What are 3 ways of controlling extracellular pathogens? 3 Marks

Three ways of controlling

- 1) Neutralization
- 2) Opsonization
- 3) Complement activation

Qno3: Differentiate between Isograft, Allograft and Xenograft. 3 Marks

Transplantation: is the process of moving cells, tissues & organs from one site to another

- One person (donor) to other person (recipient)

Isograft: Between individuals of same genetic makeup (Twins)

Allograft: One person to another non-identical person

Xenograft: across the species

Qno4: What are domains in immunoglobulin structure? 3 Marks

- 1) Heavy & Light chains:

Four chains, 02 identical light chains, 02 identical heavy chains

- 2) Disulfide bonds: Interchain & Intrachain
- 3) Variable & constant regions: based on amino acid sequences
- 4) Hinge region: makes antibodies flexible to change its shape
- 5) Oligosaccharides: attached on CH2 domain in most of immunoglobulins

Qno5: Write all steps in process of phagocytosis in details? 5 Marks

Process of Phagocytosis

1) Chemotaxis

- Chemo: Chemical, Taxis: Movement
- SOS signals from bacteria
- Secretory molecules like coagulation peptides, complement
- Migration of phagocytes across the capillary wall (Diapedesis)

2) Attachment

- Phagocytes carry receptors for binding with bacterial surface components
- Fc receptors
- Complement receptors
- Scavenger receptors
- Toll like receptors (Pattern Recognition Receptors)

3) Phagosome Formation

- Once bacteria attach with its corresponding receptor on phagocyte
- Pseudopod extends around the bacterium to form a vesicle called as Phagosome
- Phagosome contains the trapped bacteria inside

4) Phagolysosome Formation

- Phagosome with trapped bacteria fuse with secretory vesicles of phagocytes called "Lysosomes"
- Phagolysosome formation result in the action of hydrolytic enzymes i.e Lysozyme

Qno6: How the process of VDJ recombination is regulated?

Regulation

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VU Medical Zone

- Recombination Signal Sequence (RSS): Flanked B/W V, D & J exons
- Composed of conserved heptamer & nonamer separated by 2 & 1 turn signals
- Recombination occurs B/W 2 & 1 turn
- Recombination occurs after the removal of introns B/W V & J in case of light chain
- Recombination occurs after the removal of introns B/W V, D & J in case of heavy chain
- RAG1 & RAG2: Enzymes responsible for this recombination
- SCID

Qno7: Brief note on lymphocyte recirculation? 5 Marks

Lymphocyte Recirculation

- Naïve (Virgin) lymphocytes from primary lymphoid tissues move to secondary tissues
- Priming of lymphocytes
- Lymphocytes in blood enter in lymph node and percolate
- 1-2% of lymphocytes recirculate every hour

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